Solution Explanation Report

<u>Task 1 – Segmentation:</u>

From the lecture slide content, it looked like the results shown for Mean Shift segmentation would work the best for the given problem statement. That is because it is used in cases of finding features having similar color, gradients, texture – and no need to specify number of clusters.

(nir – red): this was done as due to lack of chlorophyll, buildings reflect roughly similar values for both wavelengths, making this a near zero number

Further, we threshold it (inversely) so as to binarize the image, and make the buildings 255 and surroundings 0.

<u>Task 2 – Refinement:</u>

The refinement is based on gradient descent and level sets.
The contour will be evolving to minimize an energy function penalizing zig-zag type boundaries and preventing contours from crossing building edges.

The output still had some holes.
https://learnopencv.com/filling-holes-in-an-image-using-opencv-python-c/
Using FloodFill, the holes have been removed to get the final output with an IoU value of **<u>85.26%</u>**.

<u>Task 3 – Vectorization:</u>

For vectorization, firstly the contours are extracted and approximated as polygons. Next step is to create bounding boxes for each house (i.e. get the orientation of each house). With respect to that orientation, all the contours are snapped such that they are at 90 degree angles. And finally, all the outputs are plot together as shown below: